

BIOLOGY LAST WEEK REVISION GUIDE

Strategic Exam Forecast & Syllabus Filter (2026 Boards)

Created for Thunderstudy By Pragya

STEP 0 & 1 — SYLLABUS FILTERING:

Current Syllabus Only. **DELETED TOPICS REMOVED:** Micropropagation, Tissue Culture, and Environmental Issues (Pollution) are excluded from this guide. Analysis based on sets 57/1/1, 57/5/1, 57/7/1.

STEP 2 — CHAPTER WEIGHTAGE GUIDE

Unit	High-Freq Topics	Trend
Reproduction	Menstrual Cycle, Pollination (Mutualism), ART	🔥 Very High
Genetics	Lac Operon, DNA Fingerprinting, Pedigree	🔥 Very High
Biotech	pBR322, PCR, Bt Cotton, Insulin	🔥 Very High
Welfare	Malaria Cycle, Sewage Treatment (STP)	🔥 Critical
Ecology	Logistic Curves, Species-Area Graph	High

STEP 3 — TOP 8 MOST REPEATED CONCEPTS

1. **STP (Sewage Treatment):** Identify Flocs & Activated Sludge.
2. **pBR322:** Labeling restriction sites and selectable markers.
3. **Menstrual Graph:** Identify LH surge and Progesterone peaks.
4. **Growth Curves:** Logistic (S-shape) vs Exponential.
5. **EcoRI:** Naming & Palindromes.
6. **Pedigree:** Identify Sex-linked vs Autosomal traits.
7. **Bioactive Microbes:** Cyclosporin A, Statins, etc.
8. **DNA Replication:** The Y-shaped fork mechanism.

STEP 5 — PREDICTION ZONE (Sure-Shot Marks)

🔥 5 Mark Targets:

- STP Flowchart & Biological treatment.
- pBR322 Insertional Inactivation.
- Menstrual cycle hormonal coordination.

⚡ 3 Mark Targets:

- Bt Cotton mechanism of action.
- Malaria life cycle (Host stages).
- DNA Replication Fork labels.

STEP 6 — EXPECTED PRACTICE QUESTIONS

Q1. Masses of bacteria held together by fungal filaments are:

A) Sludge B) Floccs C) Inoculum D) Scum. **Ans: B**

Q2. Hormone causing Graafian follicle rupture:

A) FSH B) Estrogen C) LH D) Progesterone. **Ans: C**

Q3. Bioactive molecule for organ transplant patients:

A) Statins B) Cyclosporin A C) Lipase D) Amylase. **Ans: B**

STEP 7 — LAST 7 DAYS QUICK MEMORY LIST

Quick Equations:

- Verhulst-Pearl: $dN/dt = rN [(K-N)/K]$
- Species-Area: $\log S = \log C + Z \log A$

Top Common Traps:

- Spermiation vs Ejaculation (Release mechanism).
- Transcription (DNA-RNA) vs Translation (RNA-Protein).
- DNA Polymerase direction is ALWAYS 5' -> 3'.

STEP 8 — FINAL DATA REPORT

Data Integrity: PDFs analyze recent 2022-2024 trends. Confidence level: Very High. Focus on STP, Biotech Principles, and Reproductive Cycles for 50%+ score guarantee.

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